Algebra 1

Quiz 2

Date: 12 October 2026 Time: 14:00–17:00 Marks: 20

- As usual explain everything fully using complete sentences.
- Unless mentioned otherwise, all vector spaces are finite-dimensional over an arbitrary field F.
- You may use any definitions and results stated and proved in class (clearly specify what result you are using), but no other definitions or results. In particular, you may not use terms or results developed exclusively in tutorials. However, you may use any standard technique of calculating determinants by clearly stating which result or property you are using.

1. Work over $\mathbb{R}$. Consider the 3×3 matrix

$$A_t = \begin{bmatrix} 1 & t & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix},$$

where t is a real parameter.

(a) Show that the vector $v = \begin{bmatrix} t \\ 1 \\ 1-t \end{bmatrix}$ is an eigenvector of A_t.

(b) Compute the characteristic polynomial of A_t, and find all values of t for which A_t is diagonalizable over $\mathbb{R}$. You do not have to actually diagonalize A_t.

(c) For value(s) of t not listed in (b) show that A_t is not diagonalizable.

2. Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ be the 2×2 Fibonacci matrix, but now considered over the finite field $\mathbb{F}_p$.

(a) Compute the characteristic polynomial $p_A(x)$ of A.

(b) Using **only** your characteristic polynomial and NO matrix computations, determine whether A is diagonalizable over each of the following fields:

$$\mathbb{F}_2 \ (p = 2) \qquad \mathbb{F}_5 \ (p = 5) \qquad \mathbb{F}_{11} \ (p = 11)$$

3. Suppose A and B are subspaces of a vector space V. The dimensions of A, B and V are 2, 7 and 10 respectively.

(a) Is it possible for A and B to span V? If yes, give an example. If not, state why. Is it possible for A and B to be independent? If yes, give an example. If not, state why.

(b) Let $A \oplus B$ be the external direct sum of A and B, i.e., $A \oplus B =$ the set $A \times B$ with addition and scalar multiplication defined slotwise. The map $f : A \oplus B \to V$ defined by $f(a, b) = a + b$ is linear. (Check this in your mind if you wish. It is not part of the question.) What is the kernel of this map?

(c) Make a table with two columns as follows. In the first column list all possible values of the dimension of the subspace $A + B$ of V. For each entry in the first column, write on the same line in the second column all possible corresponding values of the dimension of $A \cap B$. Explain.

4. Let V be a finite-dimensional vector space over an arbitrary field F, and let $T : V \to V$ be a linear operator. Suppose $\dim \ker(T) = 3$. We say that a nonzero polynomial $d(x)$ divides a polynomial $p(x)$ if $p(x) = d(x)q(x)$ for some polynomial $q(x)$.

(a) Show that x^3 divides the characteristic polynomial $p_T(x)$ of T. Hint: start with a basis of $\ker(T)$. Then complete the sentence: "By same logic, if λ is an eigenvalue of T, then _____ divides $p_T(x)$."

(b) For T satisfying the hypothesis in (a), show by a concrete example that x^4 can also divide $p_T(x)$.

(c) Does the statement in part (b) stay true if T is assumed to be diagonalizable? Justify your answer.

5. (a) Let V be an n-dimensional vector space over the finite field $\mathbb{F}_p$, where p is a prime. Let k be an integer with $0 \leq k \leq n$. Determine the total number of ordered sequences of k linearly independent vectors $(v_1, v_2, \ldots, v_k)$ in V.

(b) Find the number of k-dimensional subspaces of $\mathbb{F}_p^n$. Hint: Look at part (a). Look at it again!

Make-up problem

This problem was given on the board, separately from the quiz paper. Attempt it only to make up marks for quiz questions you did not solve. It is not extra credit, and it does not add to your quiz score otherwise.

6. Give an example of each of the following situations: an operator T on a finite-dimensional nonzero vector space V, over an infinite field F, such that the number of *nonzero proper* T-invariant subspaces of V is

(a) zero (b) one (c) two (d) infinite.

Prove all answers!